Q1 2021 (01 Sep Shift 2)

Electric field of plane electromagnetic wave propagating through a non-magnetic medium is given by $E = 20\cos\left(2 \times 10^{10} t - 200x\right) V/m$. The dielectric constant of the medium is equal to : (Take $\mu_r = 1$)

(1) 9

(2) 2

(3) $\frac{1}{3}$

(4) 3

Q2 2021 (31 Aug Shift 2)

The magnetic field vector of an electromagnetic wave is given by $B = B_o \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz - \omega t)$; where $\hat{i}, \hat{j}$ represents unit vector along x and y-axis respectively. At $t = 0$ s, two electric charges q_1 of 4π coulomb and q_2 of 2π coulomb located at $\left(0, 0, \frac{\pi}{k}\right)$ and $\left(0, 0, \frac{3\pi}{k}\right)$, respectively, have the same velocity of $0.5c\hat{i}$, (where c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is :-

(1) $2\sqrt{2} : 1$

(2) $1 : \sqrt{2}$

(3) $2 : 1$

(4) $\sqrt{2} : 1$

Q3 2021 (31 Aug Shift 1)

The electric field in an electromagnetic wave is given by $E = \left(50 NC^{-1}\right) \sin \omega(t - x/c)$

The energy contained in a cylinder of volume V is 5.5×10^{-12} J. The value of V is cm^3. (given $\epsilon_0 = 8.8 \times 10^{-12} C^2 N^{-1} m^{-2}$)

Q4 2021 (27 Aug Shift 2)

A plane electromagnetic wave with frequency of 30MHz travels in free space. At particular point in space and time, electric field is 6 V/m. The magnetic field at this point will be $x \times 10^{-8}$ T. The value of x is .....

Q5 2021 (27 Aug Shift 1)

Electric field in a plane electromagnetic wave is given by $\mathrm{E}=50 \sin \left(500 \mathrm{x}-10 \times 10^{10} \mathrm{t}\right) \mathrm{V} / \mathrm{m}$ The velocity of electromagnetic wave in this medium is :

(Given $\mathrm{C}=$ speed of light in vacuum)

(1) $\frac{3}{2} \mathrm{C}$

(2) C

(3) $\frac{2}{3} C$

(4) $\frac{\mathrm{C}}{2}$

Q6 2021 (26 Aug Shift 2)

A light beam is described by $\mathrm{E}=800 \sin \omega\left(\mathrm{t}-\frac{\mathrm{x}}{\mathrm{c}}\right)$

.An electron is allowed to move normal to the propagation of light beam with a speed of $3 \times 10^{7} \mathrm{~ms}^{-1}$. What is the maximum magnetic force exerted on the electron?

(1) $1.28 \times 10^{-18} \mathrm{~N}$

(2) $1.28 \times 10^{-21} \mathrm{~N}$

(3) $12.8 \times 10^{-17} \mathrm{~N}$

(4) $12.8 \times 10^{-18} \mathrm{~N}$

Q7 2021 (26 Aug Shift 1)

The electric field in a plane electromagnetic wave is given by

$$\overrightarrow{\mathrm{E}}=200 \cos \left[\left(\frac{0.5 \times 10^{3}}{\mathrm{~m}}\right) \mathrm{x}-\left(1.5 \times 10^{11} \frac{\mathrm{rad}}{\mathrm{s}} \times \mathrm{t}\right)\right] \frac{\mathrm{V}}{\mathrm{m}} \hat{\mathrm{j}}$$

If this wave falls normally on a perfectly reflecting surface having an area of $100 \mathrm{~cm}^{2}$. If the radiation pressure exerted by the E.M. wave on the surface during a 10 minute exposure is $\frac{\mathrm{x}}{10^{9}} \frac{\mathrm{N}}{\mathrm{m}^{2}}$. Find the value of x.

Answer Key

Q1 (1)	**Q2** (3)	**Q3** (500)	**Q4** (2)
Q5 (3)	**Q6** (4)	**Q7** (354.0)	

Q1 (1)

Speed of wave $= \frac{2\times10^{10}}{200} = 10^8 \text{ m/s}$

Refractive index $= \frac{3\times10^8}{10^8} = 3$

Now refractive index $= \sqrt{\varepsilon_r \mu_r}$

$$3 = \sqrt{\varepsilon_r(1)}$$
$$\Rightarrow \varepsilon_r = 9$$

Option (1)

Q2 (3)

$\vec{F} = q(\vec{V} \times \vec{B})$

$\dot{F}_1 = 4\pi\left[0.5\hat{i} \times B_0\left(\frac{\hat{i}+\hat{j}}{2}\right)\cos\left(K.\frac{\pi}{K} - 0\right)\right]$

$\vec{F}_2 = 2\pi\left[0.5c\hat{i} \times B_0\left(\frac{\hat{i}+\hat{j}}{2}\right)\cos\left(K\cdot\frac{3\pi}{K} - 0\right)\right]$

$\cos\pi = -1, \quad \cos 3\pi = -1$

$\therefore \frac{F_1}{F_2} = 2$

Q3 (500)

$$\mathrm{E} = 50\sin\left(\omega \mathrm{t} - \frac{\omega}{\mathrm{c}}\cdot \mathrm{x}\right)$$
$$\text{Energy density } = \frac{1}{2}\in_0 \mathrm{E}_0^2$$

Energy for volume $\mathrm{V} = \frac{1}{2}\in_0 \mathrm{E}_0^2 \cdot \mathrm{V} = 5.5\times10^{-12}$

$$\frac{1}{2}8.8\times10^{-12}\times2500\ \mathrm{V} = 5.5\times10^{-12}$$
$$\mathrm{V} = \frac{5.5\times2}{2500\times8.8} = .0005\ \mathrm{m}^3$$
$$= .0005\times10^6(\mathrm{c.m})^3$$
$$= 500(\mathrm{c.m})^3$$

Q4 (2)

$|B| = \frac{|E|}{C} = \frac{6}{3\times10^8}$

$= 2 \times 10^{-8}$ T

$\therefore x = 2$

Q5 (3)

$$V = \frac{\omega}{K} = \frac{10 \times 10^{10}}{500} = 2 \times 10^8$$
$$V = \frac{2C}{3}$$

Q6 (4)

$$\frac{E_0}{C} = B_0$$
$$F_{\max} = eB_0V$$
$$= 1.6 \times 10^{-19} \times \frac{800}{3\times10^8} \times 3 \times 10^7$$
$$= 12.8 \times 10^{-18}\ \mathrm{N}$$

Ans. (4)

Q7 (354.0)

$$E_0 = 200$$
$$I = \frac{1}{2}\varepsilon_0 E_0^2 \cdot C$$

Radiation pressure

$$P = \frac{2I}{C}$$
$$= \left(\frac{2}{C}\right)\left(\frac{1}{2}\varepsilon_0 E_0^2 C\right)$$
$$= \varepsilon_0 E_0^2$$
$$= 8.85 \times 10^{-12} \times 200^2$$
$$= 8.85 \times 10^{-8} \times 4$$
$$= \frac{354}{10^9}$$

Ans. 354.0